Power Analysis for "Can Climate Policy Build Peace?": A Conjoint Experiment in Colombia

Laura Elizabeth Moreno-Gama

Contents

1 Scope and Interpretation **1**

2 Conjoint Survey Experiment Setup **2**

3 Simulating respondents' choices in the conjoint experiment **3**

4 Estimation **7**
4.1 Numerically calibrating the DGP 10

5 Monte Carlo Simulations **12**

6 Statistical Power Under the Baseline Assumptions **17**
6.1 Choosing N for coalition heterogeneity 18

7 Sensitivity Analyses **22**
7.1 Constituency composition, model design, and within-person dependence 24

8 Minimum Detectable Effects at N = 2,000 **25**

9 Summary of Power Analysis Results **27**

10 Conclusion and Limitations **31**

1 Scope and Interpretation

I conduct a Monte Carlo power analysis to evaluate the statistical power of the proposed carbon-tax conjoint experiment in Colombia. The analysis follows the assumptions and scenarios outlined in benchmark_review.md. The effect sizes used in the simulations are informed by the literature and serve as assumptions for the power analysis. The main outcome is whether a respondent selects a given carbon-tax policy over an alternative in a paired-choice task. Therefore, the analysis estimates statistical power to detect differences in relative policy preferences, rather than absolute levels of support for carbon taxation.

The experimental design includes six paired-choice tasks per respondent, with two policy profiles presented in each task. Respondents are randomly assigned to one of two survey-order arms (A or B) and classified into four constituencies based on their attitudes toward climate policy and peacebuilding. For RQ2, I restrict the analysis to Arm A, where these attitudes are measured before respondents complete the conjoint experiment. The main simulations include the full set of conjoint attributes: revenue allocation, type of peace program, household cost, policy framing, and organizational endorsement. I also examine a reduced version of the conjoint as a sensitivity analysis to assess how the inclusion of additional attributes affects statistical power.

I evaluate statistical power across different sample sizes and assumptions about treatment effects, constituency composition, and respondent preferences. The simulations use assumed constituency proportions

and behavioral parameters rather than estimates from a pilot survey. I estimate statistical power using 500 Monte Carlo replications for each scenario and sample size, and report Monte Carlo standard errors to assess the precision of the estimates.

2 Conjoint Survey Experiment Setup

```
knitr::opts_chunk$set(echo = T, message = FALSE, warning = FALSE, results = "hide")

library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyr)
library(purrr)
library(ggplot2)
library(fixest)
library(readr)

set.seed(params$seed)
start_time <- Sys.time()

dir.create("results", showWarnings = FALSE, recursive = TRUE)
dir.create("figures", showWarnings = FALSE, recursive = TRUE)
dir.create("checkpoints", showWarnings = FALSE, recursive = TRUE)

N_grid <- c(500L, 1000L, 1500L, 2000L, 2500L, 3000L, 4000L)

group_names <- c(
  "climate_yes_peace_yes",
  "climate_yes_peace_no",
  "climate_no_peace_yes",
  "climate_no_peace_no"
)

share_levels <- c(0, 25, 50, 75)
cost_levels <- c(5000, 15000, 30000, 60000)

program_levels <- c(
  "reintegration", "victims", "crop_substitution", "rural_infrastructure"
)

frame_levels <- c(
  "carbon_tax", "climate_contribution", "climate_investment"
)
```

```r
endorser_levels <- c(
  "UN", "universities", "environmental_organization", "national_government"
)

share_targets <- matrix(
  c(0, 3, 1, -1,
    0, -5, -7, -9,
    0, 6, 4, 2,
    0, -8, -10, -12),
  nrow = 4, byrow = TRUE,
  dimnames = list(group_names, as.character(share_levels))
)

program_targets <- c(
  reintegration = 0, victims = 4,
  crop_substitution = 2, rural_infrastructure = 6
)

cost_targets <- c(
  "5000" = 0, "15000" = -2, "30000" = -5, "60000" = -10
)

frame_targets <- c(
  carbon_tax = 0, climate_contribution = 3, climate_investment = 1
)

endorser_targets <- c(
  UN = 0, universities = 3,
  environmental_organization = 1, national_government = -3
)

invisible(share_targets) # Complete benchmark matrix is defined above.
```

3 Simulating respondents' choices in the conjoint experiment

I simulate respondents' choices between two carbon-tax policy profiles using a logistic choice model. The model assumes that respondents are more likely to select the policy they prefer, based on its characteristics. For respondent i, evaluating policy j in choice task t, the probability of selecting that policy is:

$$P(Y_{itj} = 1) = \frac{\exp(U_{itj})}{\exp(U_{it1}) + \exp(U_{it2})}.$$

Here, U represents the value a respondent assigns to a policy based on its attributes. I model policy preferences as a function of the share of carbon-tax revenue allocated to peace programs, the type of peace program funded, the household cost, the policy framing, and the organization endorsing the policy. I also allow preferences to differ across respondents and constituencies. Because each respondent completes multiple conjoint tasks, the model accounts for the fact that their choices may be related across tasks. I use 0% revenue allocation to peace as the reference category and estimate the effects of allocating 25%, 50%, or 75% of carbon-tax revenue to peace programs relative to this baseline, separately by constituency. I set the model parameters so that the average effects generated by the simulations are as close as possible to the benchmark effect sizes specified in the power analysis. To do this, I evaluate the model under the full set of randomized policy attributes, using a large, balanced simulated sample and a fixed random seed to ensure

reproducibility. Finally, I specify the effects of different peace programs relative to their average effect. This allows me to distinguish the effect of allocating revenue to peace from the effect of funding a particular peace program.

```
target_table <- function(share = share_targets,
                         program = program_targets,
                         cost = cost_targets,
                         frame = frame_targets,
                         endorser = endorser_targets,
                         weights = rep(0.25, 4)) {

  share_mean <- as.numeric(crossprod(weights, share))

  tibble(
    contrast = c(
      "RQ1",
      paste0("RQ2_", seq_len(4)),
      "RQ3a_50_25", "RQ3a_75_25", "RQ3a_75_50",
      paste0("RQ3b_", names(program)[-1]),
      paste0("RQ3c_", names(cost)[-1]),
      paste0("FRAME_", names(frame)[-1]),
      paste0("ENDORSER_", names(endorser)[-1])
    ),
    target_pp = c(
      mean(share_mean[-1]),
      rowMeans(share[, -1, drop = FALSE]),
      share_mean[3] - share_mean[2],
      share_mean[4] - share_mean[2],
      share_mean[4] - share_mean[3],
      unname(program[-1] - program[1]),
      unname(cost[-1] - cost[1]),
      unname(frame[-1] - frame[1]),
      unname(endorser[-1] - endorser[1])
    )
  )
}


make_coefficients <- function(scale = 0.04,
                              share = share_targets,
                              program = program_targets,
                              cost = cost_targets,
                              frame = frame_targets,
                              endorser = endorser_targets) {

  list(
    share = scale * share,
    program = scale * (program - mean(program)),
    cost = scale * cost,
    frame = scale * frame,
    endorser = scale * endorser
  )
}
```

```
simulate_profiles <- function(N, weights = rep(0.25, 4),
                              design = "full", seed = NULL) {

  if (!is.null(seed)) set.seed(seed)

  stopifnot(
    length(weights) == 4,
    all(weights >= 0),
    abs(sum(weights) - 1) < 1e-10,
    design %in% c("core", "full")
  )

  arm <- sample(rep(c("A", "B"), length.out = N))
  group <- sample(seq_len(4), N, replace = TRUE, prob = weights)

  persons <- tibble(
    id = seq_len(N),
    arm = arm,
    group = group,
    rand_share = rnorm(N),
    rand_cost = rnorm(N)
  )

  d <- persons[rep(seq_len(N), each = 12L), ]

  d$pair <- rep(seq_len(6L * N), each = 2L)
  d$profile <- rep(1:2, 6L * N)

  d$share <- sample(share_levels, nrow(d), replace = TRUE)
  d$cost <- sample(cost_levels, nrow(d), replace = TRUE)

  d$program <- "none"

  eligible <- d$share > 0

  d$program[eligible] <- sample(
    program_levels, sum(eligible), replace = TRUE
  )

  d$frame <- if (design == "full") {
    sample(frame_levels, nrow(d), replace = TRUE)
  } else {
    rep("carbon_tax", nrow(d))
  }

  d$endorser <- if (design == "full") {
    sample(endorser_levels, nrow(d), replace = TRUE)
  } else {
    rep("UN", nrow(d))
  }

  d
}
```

```r
profile_utility <- function(d, beta, heterogeneity = 0.20) {

  share_idx <- match(d$share, share_levels)

  share_utility <- beta$share[cbind(d$group, share_idx)]

  program_utility <- numeric(nrow(d))
  eligible <- d$share > 0

  program_utility[eligible] <- beta$program[d$program[eligible]]

  share_utility + program_utility +
    beta$cost[as.character(d$cost)] +
    beta$frame[d$frame] +
    beta$endorser[d$endorser] +
    heterogeneity * d$rand_share * (d$share > 0) +
    heterogeneity * 0.5 * d$rand_cost *
      (d$cost - min(cost_levels)) / diff(range(cost_levels))
}


draw_choices <- function(d, beta, heterogeneity = 0.20,
                         seed = NULL) {

  if (!is.null(seed)) set.seed(seed)

  u <- profile_utility(d, beta, heterogeneity)

  first <- seq.int(1L, length(u), by = 2L)

  p_first <- plogis(u[first] - u[first + 1L])

  first_wins <- rbinom(length(first), 1L, p_first)

  d$chosen <- as.integer(
    rep(first_wins, each = 2L) ==
      rep(c(1L, 0L), length(first))
  )

  d$share <- factor(d$share, levels = share_levels)
  d$cost <- factor(d$cost, levels = cost_levels)
  d$program <- factor(
    d$program, levels = c("none", program_levels)
  )
  d$frame <- factor(d$frame, levels = frame_levels)
  d$endorser <- factor(d$endorser, levels = endorser_levels)

  d
}


check_design <- function(d, N) {
```

```
  stopifnot(
    nrow(d) == 12L * N,
    all(tabulate(d$id) == 12L),
    all(tabulate(d$pair) == 2L),
    all(tapply(d$chosen, d$pair, sum) == 1),
    all(as.character(d$program[d$share == "0"]) == "none"),
    all(as.character(d$program[d$share != "0"]) != "none")
  )

  invisible(TRUE)
}
```

4 Estimation

I estimate the effects of the conjoint attributes using linear probability models, with standard errors clustered at the respondent level to account for the fact that each respondent evaluates multiple policy profiles. Each model controls for the other randomized policy attributes, allowing me to estimate the effect of each attribute while accounting for the others. For RQ2, I examine whether the effect of allocating carbon-tax revenue to peace programs differs across constituencies. I estimate a single model using respondents in Arm A, interacting revenue allocation with constituency membership. This approach allows me to directly test whether the effects differ across constituencies. I estimate the effects of different peace programs using only policy profiles that allocate a positive share of revenue to peace. This ensures that comparisons between programs are based on policies that actually fund peace initiatives. Finally, I estimate the effects of policy framing and organizational endorsements using the full conjoint design, where these attributes are included.

```
contrast_test <- function(fit, weights, label) {

  b <- coef(fit)
  V <- vcov(fit)

  if (!all(names(weights) %in% names(b))) {
    return(tibble(
      contrast = label, estimate = NA_real_,
      se = NA_real_, p = NA_real_
    ))
  }

  a <- setNames(numeric(length(b)), names(b))
  a[names(weights)] <- weights

  estimate <- sum(a * b)
  se <- sqrt(max(0, as.numeric(t(a) %*% V %*% a)))

  tibble(
    contrast = label,
    estimate = estimate,
    se = se,
    p = if (se > 0) 2 * pnorm(-abs(estimate / se)) else NA_real_
  )
}


fit_conjoint <- function(d, design = "full") {
```

```
controls <- if (design == "full") {
  "cost + frame + endorser"
} else {
  "cost"
}

formula_main <- as.formula(paste("chosen ~ share +", controls))

main <- feols(
  formula_main, data = d, vcov = ~id,
  notes = FALSE, warn = FALSE
)

out <- list(
  contrast_test(
    main,
    c(share25 = 1 / 3, share50 = 1 / 3, share75 = 1 / 3),
    "RQ1"
  ),
  contrast_test(main, c(share50 = 1, share25 = -1), "RQ3a_50_25"),
  contrast_test(main, c(share75 = 1, share25 = -1), "RQ3a_75_25"),
  contrast_test(main, c(share75 = 1, share50 = -1), "RQ3a_75_50")
)

A <- d[d$arm == "A", ]
A$group_factor <- factor(A$group, levels = seq_len(4))

rq2_formula <- as.formula(
  paste("chosen ~ share * group_factor +", controls)
)

rq2 <- feols(
  rq2_formula, data = A, vcov = ~id,
  notes = FALSE, warn = FALSE
)

group_weight <- function(j) {

  w <- c(share25 = 1 / 3, share50 = 1 / 3, share75 = 1 / 3)

  if (j > 1) {
    extra <- paste0(
      "share", c(25, 50, 75), ":group_factor", j
    )

    w <- c(w, setNames(rep(1 / 3, 3), extra))
  }

  w
}

for (j in seq_len(4)) {
  out[[length(out) + 1L]] <- contrast_test(
```

```
    rq2, group_weight(j), paste0("RQ2_", j)
  )
}

pairs <- combn(seq_len(4), 2, simplify = FALSE)

for (ix in pairs) {

  x <- group_weight(ix[1])
  y <- group_weight(ix[2])

  names_all <- union(names(x), names(y))

  w <- setNames(numeric(length(names_all)), names_all)

  w[names(x)] <- w[names(x)] + x
  w[names(y)] <- w[names(y)] - y

  out[[length(out) + 1L]] <- contrast_test(
    rq2, w, paste0("RQ2_diff_", ix[1], "_", ix[2])
  )
}

positive <- d[d$share != "0", ]
positive$program <- relevel(
  droplevels(positive$program), ref = "reintegration"
)

program_formula <- as.formula(
  paste("chosen ~ program + share +", controls)
)

program_fit <- feols(
  program_formula, data = positive, vcov = ~id,
  notes = FALSE, warn = FALSE
)

for (level in program_levels[-1]) {
  out[[length(out) + 1L]] <- contrast_test(
    program_fit,
    setNames(1, paste0("program", level)),
    paste0("RQ3b_", level)
  )
}

for (level in cost_levels[-1]) {
  out[[length(out) + 1L]] <- contrast_test(
    main,
    setNames(1, paste0("cost", level)),
    paste0("RQ3c_", level)
  )
}
```

```
  if (design == "full") {

    for (level in frame_levels[-1]) {
      out[[length(out) + 1L]] <- contrast_test(
        main, setNames(1, paste0("frame", level)),
        paste0("FRAME_", level)
      )
    }

    for (level in endorser_levels[-1]) {
      out[[length(out) + 1L]] <- contrast_test(
        main, setNames(1, paste0("endorser", level)),
        paste0("ENDORSER_", level)
      )
    }
  }

  bind_rows(out)
}
```

4.1 Numerically calibrating the DGP

Calibration uses common random draws to minimize simulation noise. I target the estimands of the actual analysis, not raw utility coefficients. This step is computationally expensive but performed once for the baseline and reused for scenarios that alter only constituency weights or effect scales.

```
calibration_data <- simulate_profiles(
  params$calibration_N,
  weights = rep(0.25, 4),
  design = "full",
  seed = params$seed + 1L
)


calibration_targets <- target_table()


calibration_loss <- function(scale) {

  beta <- make_coefficients(scale = scale)

  d <- draw_choices(
    calibration_data, beta,
    heterogeneity = 0.20,
    seed = params$seed + 2L
  )

  est <- fit_conjoint(d, "full")

  comparison <- inner_join(
    est, calibration_targets, by = "contrast"
  )

  comparison <- comparison[
```

```
    !grepl("^RQ2_diff_", comparison$contrast),
  ]

  sum((100 * comparison$estimate - comparison$target_pp)^2,
      na.rm = TRUE)
}


calibration_fit <- optimize(
  calibration_loss,
  interval = c(0.005, 0.12),
  tol = 0.0005
)


calibrated_scale <- calibration_fit$minimum

beta_baseline <- make_coefficients(scale = calibrated_scale)

calibration_estimates <- fit_conjoint(
  draw_choices(
    calibration_data, beta_baseline,
    seed = params$seed + 2L
  ),
  "full"
)

calibration_report <- inner_join(
  calibration_estimates,
  calibration_targets,
  by = "contrast"
) |>
  mutate(
    realized_pp = 100 * estimate,
    deviation_pp = realized_pp - target_pp
  ) |>
  select(contrast, target_pp, realized_pp, deviation_pp)


write_csv(calibration_report, "results/calibration_report.csv")

invisible(calibration_report) # Saved in results/calibration_report.csv

cat("Calibrated common utility scale:", calibrated_scale, "\n")
cat("Maximum absolute calibration deviation (pp):",
    max(abs(calibration_report$deviation_pp), na.rm = TRUE), "\n")
```

I calibrate the data-generating process to approximate the benchmark treatment effects identified in the literature. Because the simulations use a nonlinear choice model, the realized effects may differ slightly from the assumed values. I report both the benchmark and simulated effects to assess the correspondence between the theoretical assumptions and the simulated data.

5 Monte Carlo Simulations

For each sample size and scenario, I generate simulated conjoint responses, estimate the treatment effects, and test whether they are statistically significant at the 5% level using two-sided tests. I repeat this process across independently simulated datasets and calculate statistical power as the proportion of simulations in which the null hypothesis is rejected. I account for multiple hypothesis testing using the Holm correction for the three revenue-allocation comparisons in RQ3a and the six between-constituency comparisons in RQ2. Sample size refers to the number of respondents who complete the survey, rather than the total number of policy profiles evaluated.

To manage computational time, I save simulation results every 25 replications. If an analysis is interrupted, R automatically resumes from the most recent saved results, provided the simulation assumptions remain unchanged. This also allows me to increase the number of Monte Carlo replications without repeating completed simulations.

```
simulate_once <- function(N, replication, scenario,
                          beta = beta_baseline,
                          weights = rep(0.25, 4),
                          design = "full",
                          heterogeneity = 0.20) {

  scenario_number <- sum(utf8ToInt(scenario))

  seed <- as.integer(
    (params$seed + N * 37 + replication * 1009 +
       scenario_number * 17) %% (.Machine$integer.max - 1)
  )

  d <- simulate_profiles(
    N, weights = weights, design = design, seed = seed
  )

  d <- draw_choices(
    d, beta, heterogeneity = heterogeneity,
    seed = seed + 1L
  )

  check_design(d, N)

  fit_conjoint(d, design) |>
    mutate(
      N = N,
      replication = replication,
      scenario = scenario
    )
}

checkpoint_version <- "power-analysis-checkpoints-v1"


write_checkpoint <- function(object, filename) {

  temporary <- paste0(filename, ".tmp")

  saveRDS(object, temporary)
```

```
  if (file.exists(filename)) {
    file.remove(filename)
  }

  if (!file.rename(temporary, filename)) {
    stop("Could not save checkpoint: ", filename)
  }
}


run_scenario <- function(name, Ns = N_grid,
                         reps = params$reps,
                         beta = beta_baseline,
                         weights = rep(0.25, 4),
                         design = "full",
                         heterogeneity = 0.20) {

  stopifnot(
    reps >= 1,
    params$checkpoint_batch >= 1,
    all(Ns >= 1)
  )

  scenario_dir <- file.path("checkpoints", name)

  dir.create(
    scenario_dir, showWarnings = FALSE, recursive = TRUE
  )

  result <- vector("list", length(Ns))

  for (j in seq_along(Ns)) {

    N <- Ns[j]

    filename <- file.path(
      scenario_dir, paste0("N_", N, ".rds")
    )

    specification <- list(
      version = checkpoint_version,
      scenario = name,
      N = N,
      seed = params$seed,
      beta = beta,
      weights = weights,
      design = design,
      heterogeneity = heterogeneity
    )

    completed <- list()

    if (file.exists(filename) && !isTRUE(params$reset_checkpoints)) {
```

```
    saved <- tryCatch(
      readRDS(filename),
      error = function(e) NULL
    )

    if (!is.null(saved) &&
        identical(saved$specification, specification) &&
        is.list(saved$replications)) {

      completed <- saved$replications

      cat(
        "Resuming", name, "N =", N, "from",
        length(completed), "saved replications\n"
      )

    } else {

      cat(
        "Ignoring incompatible or unreadable checkpoint:",
        filename, "\n"
      )
    }
  }

  if (length(completed) >= reps) {

    result[[j]] <- bind_rows(completed[seq_len(reps)])

    cat("Reused completed:", name, "N =", N, "\n")

    next
  }

  for (r in seq.int(length(completed) + 1L, reps)) {

    completed[[r]] <- simulate_once(
      N, r, name, beta, weights,
      design, heterogeneity
    )

    if (r %% params$checkpoint_batch == 0L || r == reps) {

      write_checkpoint(
        list(
          specification = specification,
          replications = completed
        ),
        filename
      )

      cat(
        "Saved:", name, "N =", N,
```

```
          "replications =", r, "/", reps, "\n"
        )
      }
    }

    result[[j]] <- bind_rows(completed[seq_len(reps)])
  }

  bind_rows(result)
}

power_summary <- function(raw) {

  raw |>
    group_by(scenario, N, contrast) |>
    summarise(
      reps_ok = sum(is.finite(p)),
      power = mean(p < 0.05, na.rm = TRUE),
      mean_estimate_pp = 100 * mean(estimate, na.rm = TRUE),
      .groups = "drop"
    ) |>
    mutate(
      mc_se = sqrt(power * (1 - power) / reps_ok),
      mc_low = pmax(0, power - 1.96 * mc_se),
      mc_high = pmin(1, power + 1.96 * mc_se)
    )
}


adjusted_power <- function(raw) {

  raw |>
    group_by(scenario, N, replication) |>
    group_modify(function(df, key) {

      df$p_holm <- df$p

      for (prefix in c("RQ3a_", "RQ2_diff_")) {

        ix <- grepl(paste0("^", prefix), df$contrast)

        if (any(ix)) {
          df$p_holm[ix] <- p.adjust(df$p[ix], method = "holm")
        }
      }

      df
    }) |>
    ungroup() |>
    group_by(scenario, N, contrast) |>
    summarise(
      power_holm = mean(p_holm < 0.05, na.rm = TRUE),
      .groups = "drop"
```

```
    )
}

threshold_table <- function(summary) {

  summary |>
    group_by(scenario, contrast) |>
    summarise(
      first_tested_N = if (any(power >= 0.80, na.rm = TRUE)) {
        min(N[power >= 0.80])
      } else {
        NA_integer_
      },
      .groups = "drop"
    )
}


save_scenario <- function(raw, name) {

  summary <- power_summary(raw)
  adjusted <- adjusted_power(raw)

  write_csv(summary, paste0("results/", name, "_power.csv"))
  write_csv(adjusted, paste0("results/", name, "_holm.csv"))
  write_csv(threshold_table(summary),
            paste0("results/", name, "_required_n.csv"))

  plot_data <- summary |>
    filter(
      contrast %in% c(
        "RQ1", "RQ3a_50_25", "RQ3a_75_25",
        "RQ3a_75_50", paste0("RQ2_", seq_len(4))
      )
    )

  p <- ggplot(
    plot_data, aes(N, power, color = contrast)
  ) +
    geom_line() +
    geom_point() +
    geom_hline(yintercept = 0.8, linetype = "dashed") +
    coord_cartesian(ylim = c(0, 1)) +
    theme_minimal() +
    labs(
      title = paste("Monte Carlo power:", name),
      x = "Completed respondents",
      y = "Rejection probability"
    )

  ggsave(
    paste0("figures/", name, "_power.png"),
    p, width = 10, height = 6, dpi = 200
```

```
  )

  list(summary = summary, adjusted = adjusted, plot = p)
}
```

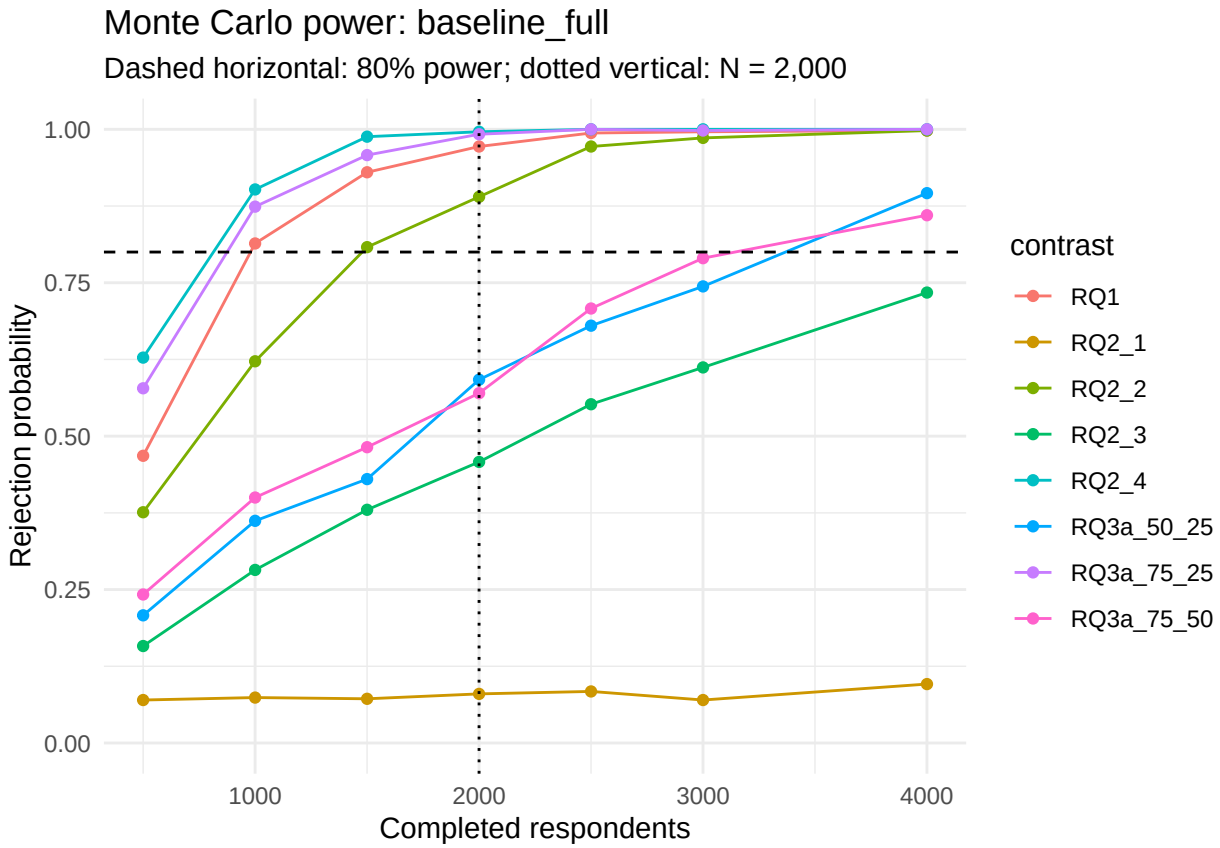
6 Statistical Power Under the Baseline Assumptions

I first estimate statistical power under the baseline assumptions about treatment effects, constituency composition, and respondent preferences. This analysis uses the full conjoint design and evaluates how statistical power changes across different sample sizes. The results provide a starting point for assessing whether the preliminary proposed sample of 2,000 respondents is sufficient to detect the effects of interest. I subsequently examine how these conclusions change under alternative assumptions in the sensitivity analyses.

```
baseline_raw <- run_scenario("baseline_full")

baseline_results <- save_scenario(
  baseline_raw, "baseline_full"
)

# Keep complete baseline power estimates in results/baseline_full_power.csv.
# Add a vertical marker at the proposed recruitment target.
print(baseline_results$plot +
  geom_vline(xintercept = 2000, linetype = "dotted") +
  labs(subtitle = "Dashed horizontal: 80% power; dotted vertical: N = 2,000"))
```



Baseline interpretation. At N = 2,000, RQ1 power is **97.2%** for a simulated mean effect of -2.95 percentage points. The four RQ2 within-constituency effects have power of **8.0%, 89.0%, 45.8%, and 99.6%**,

respectively, reflecting assumed effects of approximately +1.1, -6.85, +3.89, and -9.77 pp. The relatively low statistical power for constituency 1 reflects the small treatment effect assumed in the simulations, which is difficult to distinguish from sampling variation at the proposed sample size. Statistical power for RQ2 is further constrained by the fact that constituency-specific effects are estimated using only respondents assigned to Arm A, resulting in smaller subgroup samples and less precise estimates. Importantly, power to reject an effect *within* a group is not the same as power to reject a *difference across groups*: the latter is calculated as `RQ2_diff` and saved in the complete results. For RQ3a, power is **59.2% (50% versus 25%)**, **99.2% (75% versus 25%)**, and **57.0% (75% versus 50%)**. Hence the larger allocation contrast is well powered but the smaller adjacent-level contrasts are not. Given my primary interest in RQ1, RQ2 and RQ3, I focus the presentation of results on the average effect of peace earmarking and differences in treatment effects across constituencies. For readability, the figure omits secondary contrasts related to program type, cost, framing, and endorsements, although all estimates remain available in the saved results.

Figure labels: RQ1 is the average effect of any positive peace earmark versus 0%. RQ2_1 through RQ2_4 are *within-constituency* bundling effects, respectively: pro-climate/pro-peace, pro-climate/peace-skeptical, climate-skeptical/pro-peace, and skeptical of both. These use only survey-order Arm A. RQ3a_50_25, RQ3a_75_25, and RQ3a_75_50 compare those percentages of tax revenue earmarked for peace. The plotted powers are unadjusted: Holm-adjusted results for families of comparisons are saved separately.

6.1 Choosing N for coalition heterogeneity

The paper treats **RQ1 (average bundling)** and the **RQ2 difference between constituency 2 and constituency 3** as co-primary estimands for sample-size planning. Constituency 2 comprises respondents who support climate policy but are skeptical of peacebuilding; constituency 3 comprises respondents skeptical of climate policy but supportive of peacebuilding. Comparing their *randomized earmarking effects* provides a direct test of whether the policy bundle elicits divergent responses from the two theoretically central constituencies. This contrast is `RQ2_diff_2_3`, not either constituency's individual test against zero. Constituency membership is measured, not randomized: the causal claim concerns heterogeneous effects of randomized policy attributes, **not** the causal effect of holding a constituency attitude.

This comparison draws on the baseline Monte Carlo simulation results previously generated for the four candidate sample sizes. To improve computational efficiency, I use a checkpoint system that saves intermediate simulation results and allows completed replications to be reused rather than recomputed. The sample-size comparison therefore relies on the existing estimates of statistical power and does not require additional simulation runs. I compare unadjusted statistical power across sample sizes for the two co-primary estimands (RQ1 and the difference between constituencies 2 and 3), as well as two additional comparisons examining the effects of allocating 25% versus 50% and 50% versus 75% of carbon-tax revenues to peacebuilding. I include these latter comparisons because their relatively small assumed treatment effects make them more difficult to detect, allowing me to assess whether increasing the sample size meaningfully improves statistical power. A companion figure shows Holm-adjusted power for the constituency difference and RQ3a comparisons, because the original simulation adjusts the family of six RQ2 differences and three RQ3a contrasts separately. The RQ1 estimate is not included in those Holm families. I save the complete compact comparison as CSV rather than print another large table.

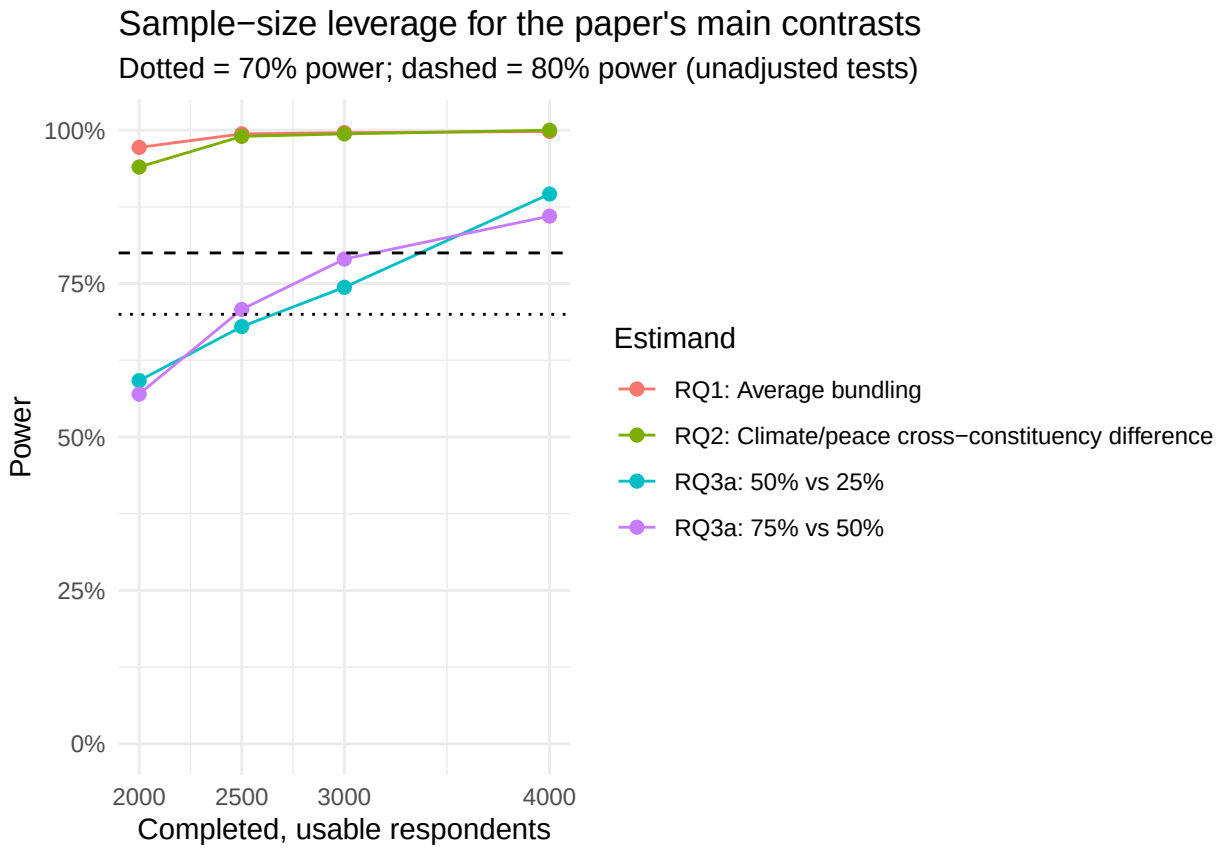
```
# Compare the SAME simulated estimands at candidate recruitment targets.
candidate_N <- c(2000L, 2500L, 3000L, 4000L)
priority_contrasts <- c("RQ1", "RQ2_diff_2_3",
                        "RQ3a_50_25", "RQ3a_75_50")
priority_labels <- c(
  RQ1 = "RQ1: Average bundling",
  RQ2_diff_2_3 = "RQ2: Climate/peace cross-constituency difference",
  RQ3a_50_25 = "RQ3a: 50% vs 25%",
  RQ3a_75_50 = "RQ3a: 75% vs 50%"
)

sample_size_leverage <- baseline_results$summary |>
```

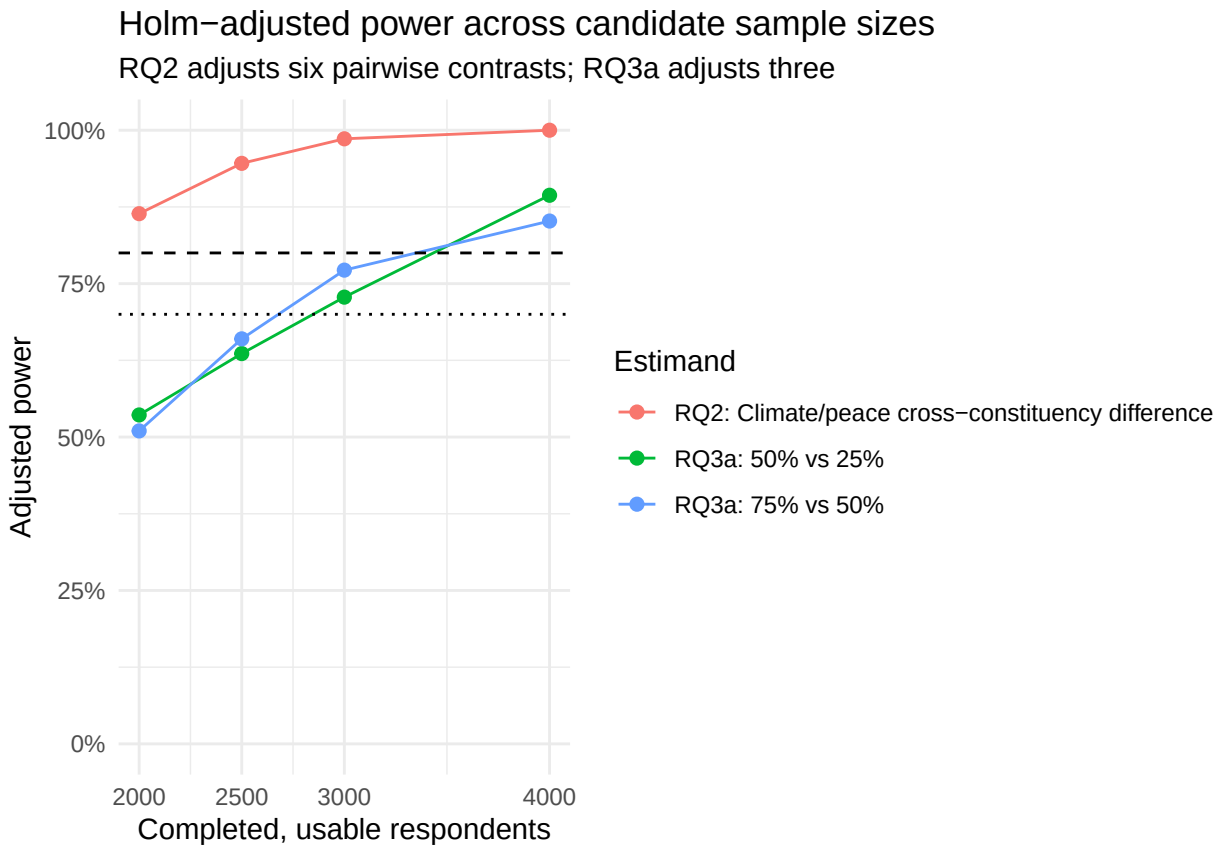
```
  dplyr::filter(N %in% candidate_N, contrast %in% priority_contrasts) |>
  dplyr::left_join(
    baseline_results$adjusted |>
      dplyr::select(N, contrast, power_holm),
    by = c("N", "contrast")
  ) |>
  dplyr::mutate(
    estimand = factor(priority_labels[contrast],
                      levels = unname(priority_labels))
  )

readr::write_csv(sample_size_leverage,
                 "results/coalition_sample_size_leverage.csv")

# Unadjusted power: common y scale makes 70% and 80% benchmarks visible.
print(
  ggplot(sample_size_leverage,
         aes(x = N, y = power, group = estimand, color = estimand)) +
    geom_line() + geom_point(size = 2) +
    geom_hline(yintercept = 0.70, linetype = "dotted") +
    geom_hline(yintercept = 0.80, linetype = "dashed") +
    scale_x_continuous(breaks = candidate_N) +
    scale_y_continuous(limits = c(0, 1), labels = scales::percent) +
    labs(title = "Sample-size leverage for the paper's main contrasts",
         subtitle = "Dotted = 70% power; dashed = 80% power (unadjusted tests)",
         x = "Completed, usable respondents", y = "Power",
         color = "Estimand") +
    theme_minimal()
)
```




```
# Adjusted power is available for RQ2 pairwise and RQ3a contrast families.
print(
  ggplot(
    sample_size_leverage |>
      dplyr::filter(contrast != "RQ1"),
    aes(x = N, y = power_holm, group = estimand, color = estimand)
  ) +
    geom_line() + geom_point(size = 2) +
    geom_hline(yintercept = 0.70, linetype = "dotted") +
    geom_hline(yintercept = 0.80, linetype = "dashed") +
    scale_x_continuous(breaks = candidate_N) +
    scale_y_continuous(limits = c(0, 1), labels = scales::percent) +
    labs(title = "Holm-adjusted power across candidate sample sizes",
         subtitle = "RQ2 adjusts six pairwise contrasts; RQ3a adjusts three",
         x = "Completed, usable respondents", y = "Adjusted power",
         color = "Estimand") +
    theme_minimal()
)
```




```
# First candidate N meeting 70% or 80%, separately for each test convention.
sample_size_thresholds <- sample_size_leverage |>
  tidyr::pivot_longer(c(power, power_holm),
                      names_to = "test", values_to = "power_value") |>
  dplyr::group_by(contrast, test) |>
  dplyr::summarise(
    first_N_70 = if (any(power_value >= 0.70, na.rm = TRUE))
      min(N[power_value >= 0.70]) else NA_integer_,
    first_N_80 = if (any(power_value >= 0.80, na.rm = TRUE))
      min(N[power_value >= 0.80]) else NA_integer_,
    .groups = "drop"
  )
readr::write_csv(sample_size_thresholds,
                 "results/coalition_sample_size_thresholds.csv")
```

Interpretation of the baseline results. For the difference between constituencies 2 and 3 (RQ2), statistical power increases from **94.0% at N = 2,000** to **99.0% at N = 2,500**, **99.4% at N = 3,000**, and **100% at N = 4,000**. Similarly, RQ1 already has high statistical power at N = 2,000 (97.2%). These results suggest that a sample of 2,000 respondents would be sufficient to detect the two main effects of interest under the baseline assumptions. However, increasing the sample size would improve statistical power for comparisons involving smaller treatment effects. For example, power for the comparison between allocating 50% and 25% of revenues to peacebuilding increases from **59.2% at N = 2,000** to **68.0% at N = 2,500** and **74.4% at N = 3,000**. Similarly, power for the comparison between allocating 75% and 50% increases from **57.0%** to **70.8%** and **79.0%**, respectively. These estimates are based on unadjusted statistical tests and the baseline simulation assumptions. I also recognize that identifying differences in treatment effects across constituencies does not necessarily imply an increase in overall support for carbon taxation, as the conjoint experiment measures preferences between alternative policy profiles.

Implications for sample size. Based on these results, I find that 2,000 respondents would provide sufficient statistical power for RQ1 and the main constituency comparison in RQ2 under the baseline assumptions. However, I would consider increasing the sample to **3,000 respondents**, as this would improve my ability to detect smaller differences in the effects of revenue allocation while also increasing the precision of the estimates. Although a sample of 4,000 would provide additional statistical power, the improvements for the main comparisons appear relatively small compared with those obtained by increasing the sample from 2,000 to 3,000.

7 Sensitivity Analyses

I conduct sensitivity analyses to examine how statistical power changes under alternative assumptions about treatment effects and constituency composition. Each scenario uses the same number of Monte Carlo replications as the baseline analysis, follows the same experimental design and estimation procedures, and saves separate tables and figures to disk, without flooding the report with output. For RQ1, I consider three alternative assumptions about the average effect of allocating carbon-tax revenue to peace programs: 0, −1, and −5 percentage points, compared with the baseline assumption of −3 percentage points. These scenarios preserve the differences in treatment effects across constituencies while changing the overall average effect, assuming that each constituency represents 25% of respondents. I also examine a scenario in which constituencies represent different proportions of the sample. Because the overall treatment effect depends on both the effects within each constituency and their relative sizes, changing constituency proportions can change the average effect even when the constituency-specific effects remain unchanged.

```
# Shift assumed treatment effects while preserving the experimental design.
shift_bundling <- function(target_pp) {

  m <- share_targets

  current <- mean(m[, -1])

  m[, -1] <- m[, -1] + (target_pp - current)

  m
}


flatten_constituencies <- function() {

  m <- share_targets

  m[, -1] <- matrix(
    rep(colMeans(share_targets[, -1]), each = 4),
    nrow = 4
  )

  m
}


alternative_constituencies <- function() {

  m <- share_targets

  m[1, -1] <- c(-1, -3, -5)
  m[4, -1] <- c(0, 0, 0)
```

```
  m
}


positive_peace_slope <- function() {

  m <- share_targets

  m[3, -1] <- c(1, 4, 7)

  m
}


scenario_specifications <- list(
  rq1_zero = list(share = shift_bundling(0)),
  rq1_minus1 = list(share = shift_bundling(-1)),
  rq1_minus5 = list(share = shift_bundling(-5)),
  weak_constituency_differences = list(
    share = flatten_constituencies()
  ),
  alternative_constituencies = list(
    share = alternative_constituencies()
  ),
  positive_peace_slope = list(
    share = positive_peace_slope()
  ),
  flat_positive_shares = list(
    share = {
      x <- share_targets
      x[, -1] <- rowMeans(x[, -1])
      x
    }
  ),
  no_program_effect = list(
    program = setNames(rep(0, 4), program_levels)
  ),
  half_cost_effect = list(cost = cost_targets * 0.5),
  double_cost_effect = list(cost = cost_targets * 2),
  no_frame_effect = list(
    frame = setNames(rep(0, 3), frame_levels)
  ),
  no_endorsement_effect = list(
    endorser = setNames(rep(0, 4), endorser_levels)
  )
)


scenario_results <- list()

for (name in names(scenario_specifications)) {

  spec <- scenario_specifications[[name]]
```

```
  beta <- make_coefficients(
    scale = calibrated_scale,
    share = if (!is.null(spec$share)) spec$share else share_targets,
    program = if (!is.null(spec$program)) spec$program else program_targets,
    cost = if (!is.null(spec$cost)) spec$cost else cost_targets,
    frame = if (!is.null(spec$frame)) spec$frame else frame_targets,
    endorser = if (!is.null(spec$endorser)) {
      spec$endorser
    } else {
      endorser_targets
    }
  )

  raw <- run_scenario(name, beta = beta)

  scenario_results[[name]] <- save_scenario(raw, name)
}
```

7.1 Constituency composition, model design, and within-person dependence

```
# Vary constituency shares, conjoint attributes, and preference heterogeneity.
structural_specs <- list(
  unequal_constituencies = list(
    weights = c(0.50, 0.20, 0.20, 0.10)
  ),
  core_design = list(design = "core"),
  low_preference_heterogeneity = list(heterogeneity = 0.05),
  high_preference_heterogeneity = list(heterogeneity = 0.50)
)


for (name in names(structural_specs)) {

  spec <- structural_specs[[name]]

  raw <- run_scenario(
    name,
    weights = if (!is.null(spec$weights)) {
      spec$weights
    } else {
      rep(0.25, 4)
    },
    design = if (!is.null(spec$design)) {
      spec$design
    } else {
      "full"
    },
    heterogeneity = if (!is.null(spec$heterogeneity)) {
      spec$heterogeneity
    } else {
      0.20
    }
```

```
  )

  scenario_results[[name]] <- save_scenario(raw, name)
}
```

Sensitivity interpretation. The sensitivity analyses show how statistical power changes under different assumptions about the size of the treatment effect and the distribution of respondents across constituencies. At N = 2,000, the baseline assumption of a −3 percentage-point effect for RQ1 yields 97.2% power. However, when I assume a smaller effect of −1 percentage point, power decreases substantially to 23.0%. Conversely, assuming a larger effect of −5 percentage points increases power to approximately 100%. When the true effect is zero, the rejection rate is 4.6%, which is close to the expected 5% significance level. These results suggest that my ability to detect the average effect of peace earmarking depends considerably on the magnitude of the effect.

I also examine how the composition of the sample affects statistical power. When I assume unequal constituency proportions (50/20/20/10 instead of 25/25/25/25), power for RQ1 decreases to 26.4%. Importantly, this reduction is not simply due to having fewer respondents in some constituencies. Changing the distribution of respondents also changes the average treatment effect implied by my assumptions, from approximately −3 to −1.04 percentage points. This suggests that the composition of the recruited sample matters for both the expected average effect and my ability to detect it. In comparison, the alternative assumptions about framing and endorsements produce relatively small changes in RQ1 power.

These findings provide an additional reason to consider increasing the sample size from 2,000 to 3,000 respondents. A larger sample would improve statistical precision and increase my ability to detect smaller treatment effects, particularly if the effects observed in the experiment are weaker than those assumed in the baseline simulations. It could also improve the precision of constituency-specific estimates, especially if some groups represent a smaller proportion of the recruited sample than expected. However, increasing the sample size would not eliminate the challenges associated with very small treatment effects or changes in constituency composition. I therefore consider 3,000 respondents a reasonable preliminary target, while recognizing that the sensitivity analyses do not yet establish the exact power gains from increasing N under each alternative scenario.

8 Minimum Detectable Effects at N = 2,000

I estimate the **minimum detectable effect (MDE)** to assess the smallest treatment effect that the proposed sample of 2,000 respondents can detect with 80% statistical power. I use the baseline assumptions about treatment effects and respondent preferences, gradually increasing or decreasing the magnitude of the assumed effects while preserving their relative differences across constituencies and policy attributes. For each treatment effect of interest, I identify the smallest magnitude considered in the simulations that achieves at least 80% statistical power.

```
# At fixed N=2000, scale allocation effects and find the first grid point at 80% power.
mde_multipliers <- seq(0, 3, by = 0.25)

mde_runs <- map_dfr(mde_multipliers, function(multiplier) {

  beta <- make_coefficients(
    scale = calibrated_scale,
    share = share_targets * multiplier,
    program = program_targets,
    cost = cost_targets,
    frame = frame_targets,
    endorser = endorser_targets
  )
```

```
  name <- paste0("mde_", gsub("\\.", "_", as.character(multiplier)))

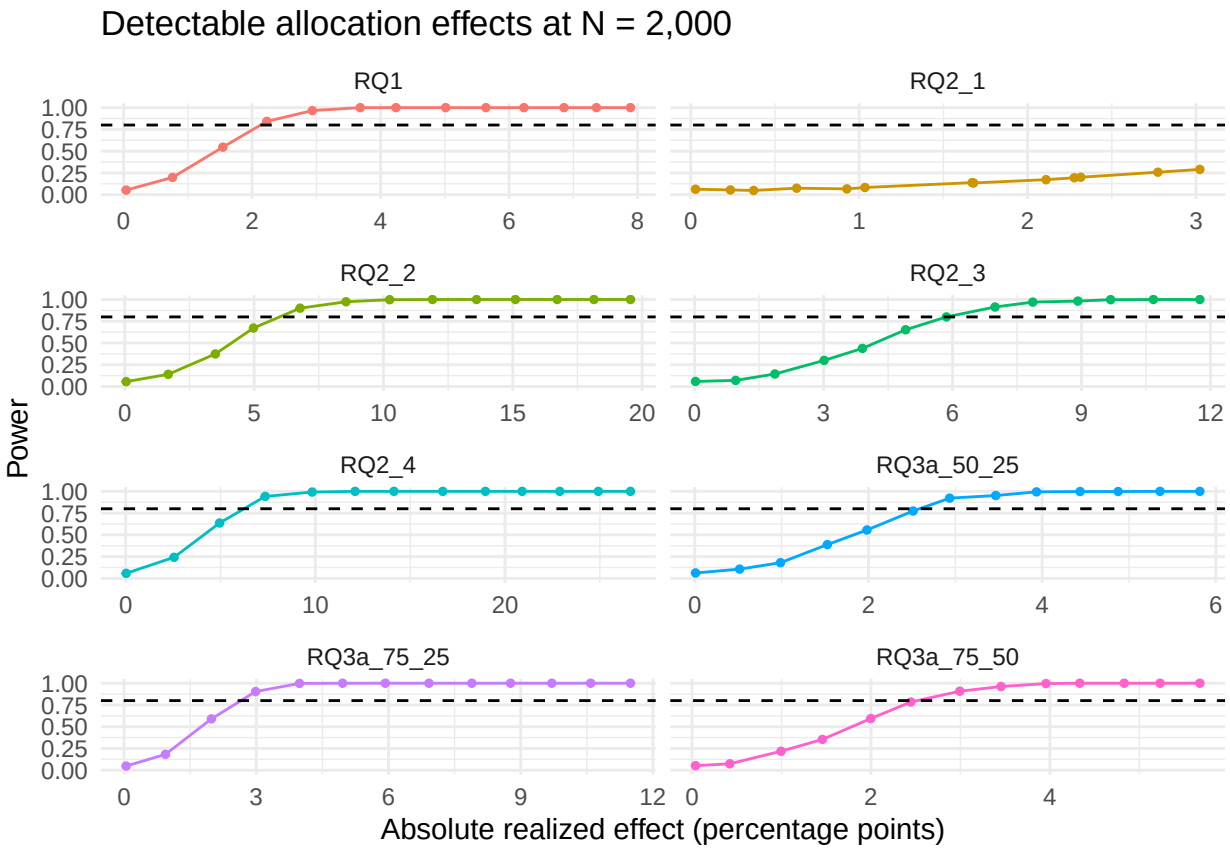
  raw <- run_scenario(
    name, Ns = 2000L,
    beta = beta
  )

  power_summary(raw) |>
    mutate(
      multiplier = multiplier,
      assumed_RQ1_pp = -3 * multiplier
    )
})


write_csv(mde_runs, "results/mde_N2000.csv")

# Report the *realized* detectable effect in percentage points, not only
# the multiplier. These are first grid crossings, not continuous optima.
mde_table <- mde_runs |>
  group_by(contrast) |>
  arrange(multiplier, .by_group = TRUE) |>
  summarise(
    first_multiplier = if (any(power >= 0.8, na.rm = TRUE))
      min(multiplier[power >= 0.8]) else NA_real_,
    detectable_effect_pp = if (any(power >= 0.8, na.rm = TRUE))
      abs(mean_estimate_pp[which(power >= 0.8)[1]]) else NA_real_,
    .groups = "drop"
  )
write_csv(mde_table, "results/mde_thresholds_N2000.csv")

# Show only allocation-related contrasts: other attribute effects are fixed.
mde_plot <- mde_runs |>
  filter(contrast %in% c("RQ1", paste0("RQ2_", 1:4),
    "RQ3a_50_25", "RQ3a_75_25", "RQ3a_75_50")) |>
  mutate(realized_magnitude_pp = abs(mean_estimate_pp))
print(ggplot(mde_plot, aes(realized_magnitude_pp, power, color = contrast)) +
  geom_line() + geom_point(size = 1) +
  geom_hline(yintercept = 0.8, linetype = "dashed") +
  facet_wrap(~contrast, scales = "free_x", ncol = 2) +
  coord_cartesian(ylim = c(0, 1)) +
  labs(title = "Detectable allocation effects at N = 2,000",
       x = "Absolute realized effect (percentage points)",
       y = "Power") +
  theme_minimal() + theme(legend.position = "none"))
```



Minimum Detectable Effects (MDEs). I also estimate the minimum treatment effects that my experiment would be able to detect with 80% statistical power, assuming a sample of 2,000 respondents. I examine these effects separately for RQ1, RQ2, and RQ3a to better understand the sensitivity of the experimental design to differences in the magnitude of treatment effects.

For RQ1, the simulations suggest that I could detect an average effect of approximately 2.25 percentage points with 80% power, compared with the 3-percentage-point effect assumed in the baseline scenario. For RQ2, the detectable effects are generally larger because I estimate constituency differences using only respondents assigned to the attitudes-first arm, which reduces the number of observations available for these comparisons. For RQ3a, I find that detecting smaller differences between revenue-allocation levels is more challenging. In particular, the comparisons between 25% and 50%, and between 50% and 75%, require effects approximately 1.5 times larger than those assumed in the baseline simulations to reach 80% power.

These results help me assess which treatment effects I can reasonably expect to detect with the proposed sample size. However, the estimates are approximate because the simulations increase or decrease all revenue-allocation effects together, rather than changing each comparison independently. In addition, the reported values correspond to the first point at which statistical power reaches 80% among the effect sizes examined, rather than an exact minimum. The calculations use unadjusted statistical tests, and the complete results are available in `results/mde_thresholds_N2000.csv`.

9 Summary of Power Analysis Results

I summarize the results from the baseline power analysis and sensitivity analyses to assess how statistical power varies across different assumptions. I focus on the proposed sample size of 2,000 respondents and compare statistical power, average estimated treatment effects, and Monte Carlo uncertainty across scenarios. This comparison provides an overall assessment of the experiment's ability to detect the effects of interest under alternative assumptions.

```r
# Load saved power analysis results
power_files <- list.files(
  "results",
  pattern = "_power\\.csv$",
  full.names = TRUE
)

# Exclude the consolidated output to avoid duplicating earlier results.
power_files <- power_files[
  basename(power_files) != "all_scenarios_power.csv"
]

# Confirm that every planned scenario has a saved result.
expected_scenarios <- c(
  "baseline_full",
  names(scenario_specifications),
  names(structural_specs)
)
found_scenarios <- sub(
  "_power\\.csv$", "", basename(power_files)
)
stopifnot(
  length(power_files) == length(expected_scenarios),
  setequal(found_scenarios, expected_scenarios)
)

# Combine results from all scenarios
all_power <- purrr::map_dfr(
  power_files,
  readr::read_csv,
  show_col_types = FALSE
)

# Save consolidated results
readr::write_csv(
  all_power,
  "results/all_scenarios_power.csv"
)

# Compare scenarios at the proposed sample size
comparison_N2000 <- all_power |>
  dplyr::filter(
    N == 2000,
    contrast %in% c(
      "RQ1",
      "RQ3a_50_25",
      "RQ3a_75_25",
      "RQ3a_75_50",
      paste0("RQ2_", 1:4)
    )
  ) |>
  dplyr::select(
    scenario,
```

```
    contrast,
    power,
    mc_se,
    mean_estimate_pp
  )

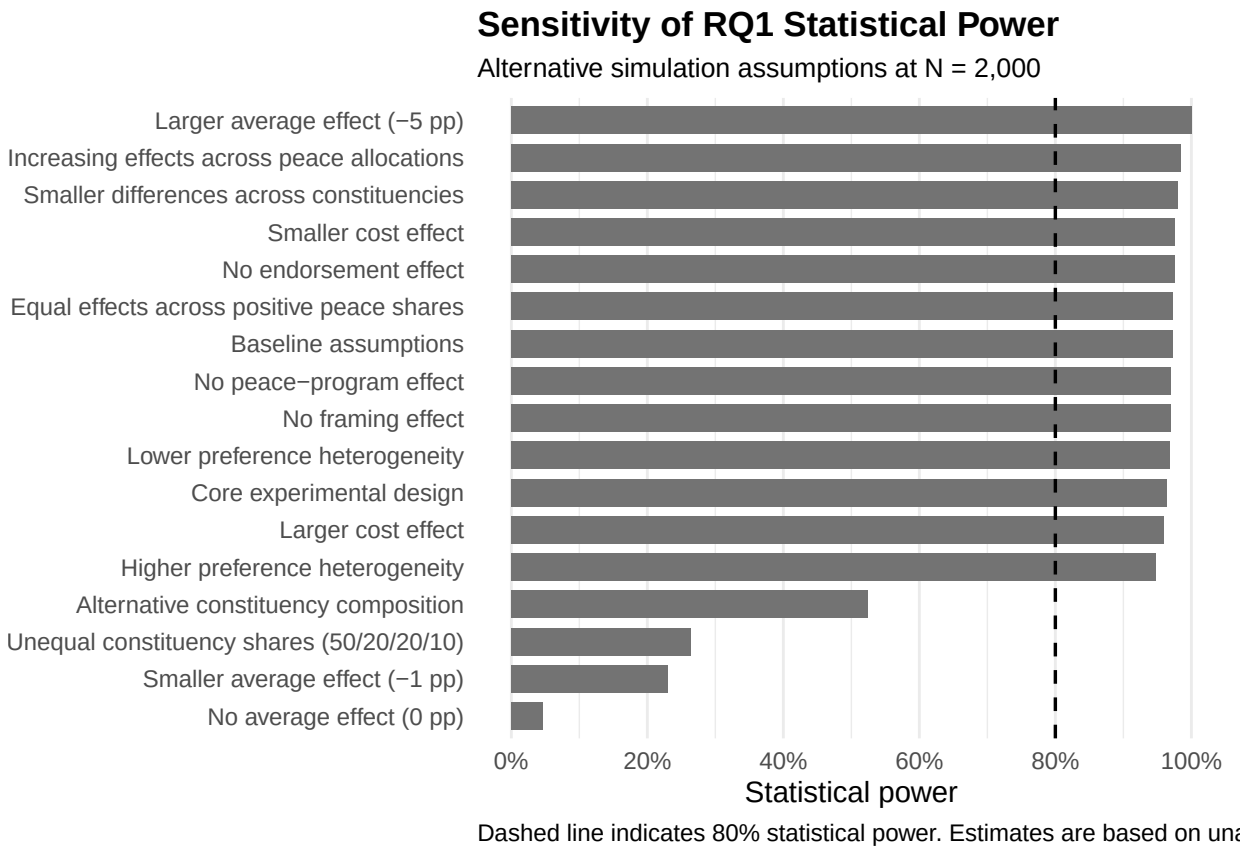
# RQ1 sensitivity analysis: statistical power across scenarios at N = 2,000

# Create readable scenario labels
scenario_labels <- c(
  "baseline_full" = "Baseline assumptions",
  "rq1_minus5" = "Larger average effect (-5 pp)",
  "rq1_minus1" = "Smaller average effect (-1 pp)",
  "rq1_zero" = "No average effect (0 pp)",
  "unequal_constituencies" = "Unequal constituency shares (50/20/20/10)",
  "alternative_constituencies" = "Alternative constituency composition",
  "positive_peace_slope" = "Increasing effects across peace allocations",
  "weak_constituency_differences" = "Smaller differences across constituencies",
  "no_endorsement_effect" = "No endorsement effect",
  "half_cost_effect" = "Smaller cost effect",
  "double_cost_effect" = "Larger cost effect",
  "flat_positive_shares" = "Equal effects across positive peace shares",
  "no_program_effect" = "No peace-program effect",
  "no_frame_effect" = "No framing effect",
  "low_preference_heterogeneity" = "Lower preference heterogeneity",
  "high_preference_heterogeneity" = "Higher preference heterogeneity",
  "core_design" = "Core experimental design"
)

# Prepare RQ1 results
rq1_summary <- comparison_N2000 |>
  dplyr::filter(contrast == "RQ1") |>
  dplyr::mutate(
    scenario_label = dplyr::recode(
      as.character(scenario),
      !!!scenario_labels
    ),
    scenario_label = reorder(scenario_label, power)
  )

# Plot sensitivity analysis
print(
  ggplot2::ggplot(
    rq1_summary,
    ggplot2::aes(x = scenario_label, y = power)
  ) +
    ggplot2::geom_col(fill = "grey45", width = 0.75) +
    ggplot2::geom_hline(
      yintercept = 0.80,
      linetype = "dashed",
      linewidth = 0.6
    ) +
    ggplot2::coord_flip(ylim = c(0, 1)) +
```

```
    ggplot2::scale_y_continuous(
      breaks = seq(0, 1, 0.2),
      labels = scales::percent_format(accuracy = 1)
    ) +
    ggplot2::labs(
      title = "Sensitivity of RQ1 Statistical Power",
      subtitle = "Alternative simulation assumptions at N = 2,000",
      x = NULL,
      y = "Statistical power",
      caption = paste(
        "Dashed line indicates 80% statistical power.",
        "Estimates are based on unadjusted tests."
      )
    ) +
    ggplot2::theme_minimal(base_size = 11) +
    ggplot2::theme(
      plot.title = ggplot2::element_text(face = "bold"),
      plot.subtitle = ggplot2::element_text(size = 10),
      plot.caption = ggplot2::element_text(
        size = 9,
        hjust = 0
      ),
      panel.grid.major.y = ggplot2::element_blank()
    )
)
```



```r
# Save the complete comparison table
readr::write_csv(
  comparison_N2000,
  "results/scenario_comparison_N2000.csv"
)
```

Reading the scenario comparison. This figure shows how statistical power for RQ1 changes under different assumptions about treatment effects, respondent preferences, and constituency composition, holding the sample size constant at 2,000 respondents. The scenarios represent alternative assumptions used in the simulations, rather than different experimental treatments. Each bar indicates the probability of detecting a statistically significant average effect of peace earmarking under the corresponding scenario, while the dashed line represents the conventional 80% statistical power threshold.

I find that statistical power remains relatively stable across several alternative assumptions, including changes in framing, endorsements, program effects, and the cost attribute. However, power decreases substantially when I assume smaller treatment effects or different constituency distributions. For example, power decreases from 97.2% under the baseline assumptions to 23.0% when the average effect is reduced to -1 percentage point, and to 26.4% when I assume unequal constituency proportions (50/20/20/10). Importantly, changing constituency proportions also changes the average treatment effect implied by the simulations, rather than simply reducing statistical precision.

These findings suggest that while 2,000 respondents would provide high statistical power under the baseline assumptions, this sample size may be insufficient if the true treatment effects are smaller or the recruited sample differs substantially from the expected constituency distribution. This provides an additional reason for me to consider increasing the sample size to 3,000 respondents. The complete scenario-by-contrast comparison, including RQ2 and RQ3a, is available in `results/scenario_comparison_N2000.csv`.

10 Conclusion and Limitations

This power analysis evaluates the sample size needed to examine how allocating carbon-tax revenues to peacebuilding programs affects policy preferences in Colombia. I am particularly interested in whether combining climate and peacebuilding objectives can generate support across political constituencies with different policy preferences. To examine this question, I focus primarily on the average effect of allocating a positive share of carbon-tax revenues to peacebuilding (RQ1) and differences in this effect across four constituencies defined by their attitudes toward climate policy and peacebuilding (RQ2). Within RQ2, I am especially interested in comparing respondents who support climate policy but are skeptical of peacebuilding with those who support peacebuilding but are skeptical of climate policy. This comparison is particularly relevant for understanding whether combining these policy objectives produces different responses among constituencies whose preferences do not necessarily align.

Under the baseline assumptions, I find that 2,000 completed surveys provide 97.2% statistical power for RQ1 and 94.0% unadjusted power for the main constituency comparison in RQ2. These results suggest that the proposed sample would be sufficient to detect the two main effects of interest if the true treatment effects are reasonably close to those assumed in the simulations. However, statistical power varies across the other comparisons. In particular, I find that smaller differences between revenue-allocation levels (RQ3a) are more difficult to detect, as are some of the differences between constituencies.

Increasing the sample size to 3,000 respondents would improve statistical power for several of these comparisons. For example, power for the comparison between allocating 50% and 25% of revenues to peacebuilding increases from 59.2% at N = 2,000 to 74.4% at N = 3,000, while power for the comparison between 75% and 50% increases from 57.0% to 79.0%. These improvements are relevant because I am interested not only in whether allocating revenues to peacebuilding affects policy preferences, but also in whether the size of that allocation matters. A larger sample would also improve the precision of constituency-specific estimates, particularly for comparisons involving smaller treatment effects.

Based on these findings, I consider 3,000 completed surveys a reasonable preliminary sample-size target.

Although 2,000 respondents would provide high power for my two main comparisons under the baseline assumptions, increasing the sample would allow me to examine additional differences across constituencies and revenue-allocation arrangements with greater precision. I also recognize that a larger sample cannot guarantee sufficient power for every comparison, particularly when the underlying treatment effects are very small.

Several assumptions are important when interpreting these findings. First, the treatment effects used in the simulations are informed by existing research rather than estimates from a pilot study. I calibrate the simulation model to approximate these benchmark effects, but the calibration adjusts a common scale across model parameters rather than matching each effect individually. As a result, the treatment effects generated by the simulations may differ somewhat from the original assumptions. The sensitivity analyses help me assess how statistical power changes under alternative assumptions, although changes in model parameters do not necessarily correspond to identical changes in percentage-point effects. I also report Monte Carlo standard errors to account for uncertainty in the estimated power, and I interpret small differences between simulation results cautiously.

Second, the composition of the recruited sample may differ from the baseline assumptions. The main simulations assume that the four climate and peace constituencies are equally represented, while the sensitivity analyses also consider unequal proportions of 50%, 20%, 20%, and 10%. These alternative distributions can affect both the precision of constituency-specific estimates and the average treatment effect implied by the model. This is particularly important for RQ2, where I estimate differences across constituencies using only respondents assigned to Arm A (attitudes-first), resulting in smaller subgroup samples. Although the experiment includes two survey-order arms, the baseline simulations assume similar treatment effects across them and do not explicitly model differences arising from survey order.

Third, the simulations represent respondents' choices using a single model that incorporates all conjoint attributes and allows the effects of revenue allocation to vary across constituencies. In addition to RQ1, RQ2, and RQ3a, I examine the effects of peace-program type (RQ3b), household costs, policy framing, and organizational endorsements. I account for multiple hypothesis testing using the Holm correction for the six pairwise constituency comparisons in RQ2 and the three revenue-allocation comparisons in RQ3a.

Finally, the simulations assume that respondents complete all six conjoint tasks and provide usable responses. They do not explicitly model survey attrition, incomplete responses, or exclusions following data-quality checks. Therefore, the sample sizes considered refer to completed and usable surveys, rather than the total number of individuals recruited. Similarly, although I may consider oversampling respondents from conflict-affected areas, the current simulations do not explicitly model geographic recruitment or survey weights. The alternative constituency distributions provide an initial assessment of how sample composition matters, but they should not be interpreted as simulations of conflict-area oversampling. If geographic differences become central to the sampling strategy, I would need to incorporate them more explicitly into the power analysis.

Overall, I view these results as a basis for determining an appropriate recruitment target rather than as a definitive prediction of the effects the experiment will uncover. The simulations suggest that my main research questions can be examined with a sample of 2,000 respondents under the baseline assumptions, while increasing the sample to 3,000 would provide greater precision for additional comparisons.